

The divergent channels

The chinese name is Jing bie which means “divergencies from the main channels”

They run deeper than the primary vessels, starting at the lower end of the main channels, connecting with the internal organs and arising to the neck.

They are distributed on the chest, abdomen and head mainly and serve as extensions of the main meridians and supplement where the main pathway does not reach

They govern the inside of the body, running between the zang and fu, so don't have points

The yang divergents follow this pathway:

They arise from their associated channel, usually on the limbs, enter the yang organ, and often the yin organ, then rejoin the main channel at the neck at the supra clavicular fossa

The yin divergents follow this pathway:

They leave their associated channel on the limbs

The heart, lung and kidney channels enter their associated organs,

They join their paired divergent channel, then

They join the yang paired main channel at the neck

In this way, they regulate qi to and from the head and link most of the yin and yang organs.

They do not have their own points, but they have points where the channels meet and then exit.

UB/KID - UB10

GB/LIV - GB1

ST/SP - ST1

SI/HE - UB1

SJ/PE - SJ16

LI/LU - LI18

Functions and clinical indications:

- balance the yin and yang in the interior of the body and in the internal organs - **strengthen the connection between the yin and yang organs**
 - balance the flow of **yin and yang between the head and the body** - especially balance ascending and descending of the yin and the yang.
- 3 of the 6 meeting points are window of the sky points which specifically open the connection between the head and the body
- connect and integrate the organs internal organs - the yang divergents flow between the yin and the yang organs, so strengthening the connections.
 - Like connecting channels, but the connecting channels connect and integrate and balance mainly in the limbs, whereas the divergents do this in the deeper body
 - do not have their own points or pathologies

- treated via their meeting points or the points on their respective channels
- they provide an explanation for why certain points have certain functions
- UB57 for the anus for example
- The connecting points connect yin yang pairs of channels at a superficial level and in the limbs. The meeting points of the divergents connect the yin pairs of channels in the interior, at a deep level.
- they also connect the yin -yang pairs of organs
- we can use yang points to treat yin problems and vice versa GB divergent goes to the heart, therefore the use of GB40 can treat timidity, insomnia, difficulty making decisions and anxiety.
- All 3 yang channels go to the heart, so they provide the heart with yang qi to pump the blood and also fuel the mind

Pairs of channels and connection between yin and yang

UB and KID

- divergent channels of kid and Ub strengthen the connection between these 2 channels
- both originate behind the knee close to UB40, so in this way UB40 activates both kidney and UB channel in the case of back pain
- UB10 - meeting point - regulates the flow of UB qi and KID qi between the body to the head, so can bring yang qi down or kidney qi up to the head
- UB channel flows through the heart, so it strengthens the function of heart and kidney connection

GB and LIV

- meeting point is GB1, reinforcing the function of the liver with the eyes and the head
- The liver channel does not flow to the head or the eyes but the divergent connection means that the liver's area of influence is expanded
- There is a lower meeting point of these 2 divergent channels at the pubis, which means that the effect of both the liver and GB channels on genital issues, especially with dampness, is explained

ST and SP

- The stomach divergent channel flows through the spleen
- The upper meeting point is at ST1 and the lower is on the upper thigh close to ST30 - disharmony of stomach and spleen

SI and Heart

- upper connection is UB1 so explains the connection do the heart with the inner canthus
- The divergent channels of SI and heart meet here therefore reinforce the

connection of yin and yang of the eyes - insomnia or narcolepsy

- The lower meeting point is in the axilla. the heart divergent goes from the main channel in the axilla to the heart itself then to the SI divergent in the neck - radiation of heart pain to scapula, upper back and neck. Also the heart divergent strengthens the connection between the heart and emotional stress and the throat and communication

SJ and Pericardium

- upper meeting point is at SJ16 - window of the sky
- effect of pericardium channel points on the lateral neck and ear and lateral side of the head
- emotional stress and liver yang rising - SJ16 acts to regulate too much yang in the head

LI and lung

- upper meeting is LI18 – window of the sky
- both divergent channels flow through each other's primary and paired organs, so this pair are totally integrated
- explains why you can use lung 7 and LI4 for head and neck issues
- as the meeting point is a window of the sky it regulates yin and yang in the head - LI4 for headaches

KEY POINTS ABOUT MUSCLE CHANNELS

The muscle channels in TCM are not just muscles and they are not really channels as they have no points. They contribute to the overall circulation of qi and blood and support the frame of the body.

In summary:

They broadly follow the course of the main meridians (except UB)

They are on the surface of the body

They do not connect with the internal organs

They all originate at the extremities

They fall in to yin/yang/ arm/leg groups - 4

They broadly follow the contours of the major muscle groups spanning more than one muscle or just more than one channels

The trajectories of the 12 primary channels and connections

a) LU - primary

originates in the middle burner, near the stomach

descends to the large intestine - connection here between the yin and yang organs – internal/external

comes back up via the stomach to the diaphragm and passes through the diaphragm to enter both lungs. Then ascends to the throat, then follows the clavicle to enter the axilla before it emerges at LU 1 - the first point - ascends to

lung 2 in the middle of the hollow of the delto-pectoral triangle then descends along the antero-lateral aspect of the upper arm - runs lateral to both the pericardium and the heart channels. Descends down the forearm, down by the lateral or radial border of the radial artery, traverses the thenar eminence, down the thumb to terminate at the radial side of the thumbnail at LU11.

A branch leaves the main pathway proximal to the wrist, passes around the dorsum of the hand and passes down the index finger to the tip.

The lung channel connects with the following organs:

ST LI LUNG - therefore can treat disorders of all three organs

MEETING POINTS:

NONE

b) LI – primary

Begins at the radial side of the tip of the index finger, between the first and second metacarpals, between ext pollicis longus and brevis at the wrist. Runs up the lateral aspect of the forearm to the lateral aspect of the elbow, up to the anterior aspect of the shoulder and over the shoulder joint.

Intersects SI12, then DU14, where it meets the other 5 yang channels, then runs to the supra clavicular fossa down to the lung - making a connection with its sister channel - through the diaphragm to the large intestine fu.

Another branch arises from ST12 in the SCF, up the lateral aspect of the neck to the cheek and lower gum (hence this channel can be used to affect tooth pain or dental issues/face pain.)

Curves around the lips, then intersects with ST4 and crosses the philtrum to end at LI20

At ST12, another branch leaves the LI channel, goes via the lung, through the diaphragm, via the LI and to the lower limb to ST 37 - the lower He Sea point of the LI -This is where the functions of the LI can also be accessed.

because of where the channel runs, gace, lips, teeth, gums forehead, eyes and nose can all be treated by the LI channel. As it is a yang channel, it is often used to help clear pathogenic factors and is used for extreme heat conditions.

The LI channel connects with the following organs:

LI and lung

Meeting Points:

ST4, ST12, SI12, DU14, DU26, REN24

c) ST – primary

Begins at the lateral side of the nose at Yingxiang L.I.-20

Ascends to the medial canthus where it meets the Bladder channel at Jingming BL-I,

Descends laterally along the infra-orbital ridge to Chengqi ST-1, where it emerges. Descends to enter the upper gum and curves to meet with Yinjiao DU-28 and Renzhong DU-26 circles around the lips and meets the Conception vessel at Chengjiang REN-24 in the mento-labial groove of the chin runs laterally across the cheeks to Daying ST-5 and to Jiache ST-6 at the angle of the mandible, ascends anterior to the ear passing via Xiaguan ST-7 to Shangguan GB-3 ascends within the hairline of the temporal region to Touwei ST-8, passing via Xuanli GB-6, Xuanlu GB-5 and Hanyan GB-4, follows the hairline to meet the Governing channel at Shenting DU-24.

A branch

- separates at ST-5 and descends along the anterior border of the sternocleidomastoid muscle in the throat region at ST9, to enter the supraclavicular fossa at ST-12 travels posteriorly to the upper back where it meets the Governing channel at Dazhui DU-14 descends internally, through the diaphragm, linking with Shangwan REN-13 and Zhongwan REN-12 to enter the Stomach and connect with the Spleen.

A further branch

descends from Quepen ST-12 along the mammillary line, 4 cun lateral to the midline as far as Rugen ST-18, then passes 2 cun lateral to the midline and descends alongside the umbilicus to Qichong ST-30 in the inguinal region.

A further branch

originates from the pyloric orifice of the Stomach, descends within the abdomen and meets with the previous portion of the channel at Qichong ST-30. From the inguinal region at Qichong ST-30 the channel Travels laterally to Biguan ST-31 on the antero-lateral aspect of the thigh, descends along the lateral margin of the femur to the patella and alongside the lateral margin of the tibia to the dorsum of the foot, terminating at the lateral side of the tip of the second toe at Lidui ST-45.

A further branch

separates from the main channel at Zusanli ST-36, three cun below the knee, and runs parallel to the main branch to terminate at the lateral aspect of the middle toe.

A further branch

separates on the dorsum of the foot at Chongyang ST-42 and terminates at the medial side of the tip of the big toe at Yinbai SP-1 where it links with the Spleen channel.

The stomach channel meets with the following organs:

Stomach and spleen

Meeting points:

LI20, UB1, GB3, 4, 5, 6, DU14, 24, 26, 28, REN12, 13, 24

d) SP – primary

begins at the medial side of the tip of the big toe at Yinbai SP-I

runs along the medial aspect of the foot, following the border where the skin changes color

ascends in front of the medial malleolus at Shangqiu SP-5

follows the posterior border of the tibia up the medial aspect of the leg to a point **8 cun superior to the medial malleolus where it crosses (and then travels anterior to) the Liver channel,**

ascends along the medial aspect of the knee and the antero-medial aspect of the thigh to the lower abdomen where it intersects the Conception vessel at Zhongji **REN-3, Guanyuan REN-4 and Xiawan, REN-10**, before entering the Spleen and connecting with the Stomach,

emerges in the region of the Stomach and ascends first at 4 cun lateral to the midline then at 6 cun lateral to the midline, passing through Riyue **GB-24, Qimen LIV-14 and Zhongfu LU-1**, and descends to terminate in the seventh intercostal space on the mid-axillary line at Dabao SP-21.

A branch

- ascends through the diaphragm, runs alongside the esophagus and spreads over the lower surface of the tongue – spleen opens into tongue

A further branch

- ascends from the Stomach, passes through the diaphragm and flows to link with the Heart.

Spleen channel connects with the following organs:

spleen, stomach and heart

Meeting points:

LU1, GB24, LIV14, REN3, REN4, REN 10, REN17

e) HE - primary

Originates in the Heart

Emerges from the system of blood vessels surrounding the Heart and descends through the diaphragm to connect with the Small Intestine,

One branch separates from the Heart, ascends alongside the esophagus and then across the face and cheek to connect with the tissues surrounding the eye – think inner canthus

Another branch travels directly from the Heart to the Lung and descends to emerge from the axilla at Jiquan HE-1, then travels along the medial aspect of the upper arm (medial to the Lung and Pericardium channels) to the elbow at Shaohai HE-3, descends along the antero-medial aspect of the lower arm to the pisiform bone of the wrist at Shenmen HE-7, travels through the palm and along the radial side of the little finger to terminate at the radial corner of the nail at Shaochong HE-9.

The heart channel connects with the following organs:

Heart, Lung and SI

There are no meeting points

f) SI – primary

originates at the ulnar side of the tip of the little finger at Shaoze SI-1, ascends along the ulnar aspect of the hand to reach the wrist where it emerges at the styloid process of the ulna at Yanglao SI-6, follows the ulna to the medial aspect of the elbow, where it passes between the olecranon of the ulna and the medial epicondyle of the humerus at Xiaohai SI-8, runs along the posterior aspect of the upper arm (intersecting the Large Intestine channel at Binao L.I-14) to the posterior aspect of the shoulder joint at Naoshu SI-10, zig-zags from the inferior fossa to the superior fossa of the scapula through Tianzong SI-11 and Bingfeng SI-12, and then to the medial aspect of the scapular spine at Tianjing SI-13, crosses via Jianwaishu SI-14 and Jianzhongshu SI-15 to Dazhui DU-14 at the lower border of the spinous process of C7, intersecting the Bladder channel at Fufen BL-41 and Dazhu BL-11, descends into the supraclavicular fossa at Quepen ST-12 and **connects with the Heart**, descends along the oesophagus, intersects the Conception vessel at Shanzhong REN-17 and passes through the diaphragm to the Stomach, intersects the Conception vessel at Shangwan REN-13 and Zhongwan REN-12 and enters the Small Intestine.

A branch

ascends from the supraclavicular fossa to cross the neck and cheek to the outer canthus of the eye, where it meets the Gall Bladder channel at Tongziliao GB-1, then travels posteriorly towards the ear, where it intersects the Gall Bladder channel at

Touqiaoyin GB-11 and the Sanjiao channel at Jiaosun SJ-20 and Erheliao SJ-22

and enters the ear at Tinggong SI-19.

Another branch

separates from the previous branch on the cheek and ascends to the infra-orbital region (Quanliao SI-18) then along the lateral aspect of the nose to the inner canthus where it meets with the Bladder channel at Jingming BL-I.

According to the Spiritual Pivot another branch descends from the SI fu to Xiajuxu ST-39, the lower he-sea point of the Small Intestine.

The SI primary channel connects with the following organs:

Heart, Stomach and SI

Meeting points:

LI14, DU14, UB11, UB41, ST12, REN17, 12, 13, GB1, 11, SJ20, 22, UB1

g) UB – primary

begins at the inner canthus of the eye at Jingming BL-I and ascends along the forehead towards the vertex, firstly to intersect with Toulinqi GB-15, Shenting DU-24 and then Baihui DU-20,

from the vertex, a branch descends to the temples in the region above the ear, intersecting the Gall Bladder channel at points Qubin GB-7, Shuaigu GB-8, Tianchong GB-9, Fubai GB-10, Touqiaoyin GB-11 and Wangu GB-12, from the vertex, another branch enters the brain, then meets Naohu DU-17, and then emerges to descend to the nape of the neck where the channel splits into two branches.

The first (medial) branch

descends along the posterior aspect of the neck, intersecting Dazhui DU-14 and Taodao DU-13, then descends alongside the spine, 1.5 cun lateral to the midline, to the lumbar region,

penetrates deep into the interior via the paravertebral muscles to connect with the Kidneys and link with the Bladder,

a sub-branch separates in the lumbar region, descends along the sacrum, crosses the buttock and descends to the popliteal fossa of the knee at Weizhong BL-40.

The second (lateral) branch

separates at the nape of the neck and descends to the medial border of the scapula and then parallel to the spine, 3 cun lateral to the midline, to the gluteal region,

crosses the buttock to intersect at Huantiao GB-30, then descends along the postero-lateral aspect of the thigh to meet with the previous branch of the channel in the popliteal fossa at Weizhong BL-40,

descends through the gastrocnemius muscle, emerges posterior to the lateral

malleolus at Kunlun BL-60, then follows along the fifth metatarsal bone to terminate at Zhiyin BL-67 at the lateral side of the tip of the fifth toe, where it meets with the Kidney channel.

The UB channels connects with the following organs:

Kidney and UB

Meeting points:

DU20, 24, GB8, 9,10, 11, 12, 15, DU13, 14, 17, GB30

h) KID – primary

Begins below the little toe (UB67), Crosses the sole of the foot to Kid 1, ascends to Kid 2, anterior and inferior to the navicular tuberosity.

Travels to Kid 3, posterior to the medial malleolus, then descends to the heel then across to Kid 6, just below the medial malleolus.

Ascends the medial aspect of the leg, meets the **spleen channel at Spleen 6**, ascends to the medial aspect of the popliteal crease at Kid 10.

Ascends along the postero-medial aspect of the thigh to the tip of the coccyx and **meets the DU channel at Du 1**

Threads through the spine and enters the kidney and connects with the Urinary Bladder. Connects with the **Ren channel at Ren 3, Ren 4 and Ren 7**.

One branch emerges from the kidney, ascends through the liver, the diaphragm, enters the lung and ascends to the throat to terminate at the **root of the tongue**.

Another branch separates from the lung, connects with the heart, disperses into the chest and connects with the pericardium channel and Ren 17

The Kidney channel connects with the following organs:

Kid, UB, Liv, Lu, Heart

Meeting Points:

SP6, DU1, REN3, 4, 7, 17

i) PE - primary

Originates in the center of the chest connects with the pericardium and descends through the diaphragm to the abdomen – passes through all 3 jiao

One branch:

Runs from inside the chest and emerges just by the nipple, around 3 cun inferior to the anterior axillary fold (PE1)

Arches over the axilla, follows the antero-medial aspect of the upper arm between the lung and the heart channels to the cubital crease (PE3)

Descends the forearm between the tendons of palmaris longus and flexor carpi radialis to the palm (PE8)

Travels along the palm to the tip of the middle finger (PE9)

Another branch:

Arises from the palm at PE8 and follows the radial aspect of the ring finger to its tip (close to SJ1)

The Pericardium channel connects with the following organs:

San Jiao, pericardium

There are no meeting points

j) SJ – primary

begins at the ulnar aspect of the tip of the ring finger and runs between the fourth and fifth metacarpal bones along the dorsum of the hand, travels up the posterior aspect of the forearm between the radius and the ulna and between the Large and Small Intestine channels, traverses the olecranon of the ulna at Tianjing SJ-10 and continues up the postero-lateral aspect of the upper arm to the shoulder where it intersects the Small Intestine channel at Bingfeng SI-12, travels towards the spine via Dazhu BL-11 where it intersects the Governing vessel at Dazhui DU-14, ascends laterally to the highest point of the shoulder where it intersects the Gall Bladder channel at Jianjing GB-21, descends anteriorly into the supraclavicular fossa at Quepen ST-12, then disperses midway between the breasts at Shanzhong REN-17, connects with the Pericardium then descends through the diaphragm to the abdomen via Zhongwan REN-12, linking along its pathway the upper, middle and lower jiao.

A branch of the channel:

- separates in the region of Shanzhong REN-17,
- ascends to emerge from the supraclavicular fossa,
- rises along the neck to the posterior aspect of the ear,
- circles behind the ear via Touqiaoyin GB-11 to the temples where it intersects the Gall Bladder channel at Xuanli GB-6, Xuanlu GB-5, Hanyan GB-4 and Yangbai GB-14,
- winds down across the cheek, intersecting the SI channel at Quanliao SI-18,
- ascends to the inferior aspect of the eye.

Another branch:

- separates behind the ear and enters the ear,
- emerges in front of the ear to intersect the Small Intestine and Gall Bladder channels at Tinggong SI-19 and Shangguan GB-3,

- crosses the previous branch on the cheek to terminate at the outer canthus of the eye at Sizhukong SJ-23, linking with Tongziliao GB-1.

According to the Spiritual Pivot a branch of the Sanjiao primary channel descends to Weiyang BL-39.

The San Jiao channel connects with the following organs:

San Jiao (all 3 jiao) Pericardium

Meeting points:

SI12, 18, 19, UB11, DU14, GB21, ST12, REN12, 17, GB1, 3, 4, 5, 6, 11, 20, 21,

k) GB – primary

begins at the outer canthus by the eye. It crosses to the ear and then ascends to the upper border of the zygomatic arch and to the corner of the forehead. It then descends to the region of the ear around San jiao 22 and then curves around the ear parallel to the San Jiao channel to the mastoid process. It curves towards the side of the head to the corner of the forehead at stomach 8 and descends to just above the orbital bone at GB14. It then curves across the side of the head to just below the occiput at GB 20 and then crosses to the top of the shoulder at GB 21. It meets all the other yang channels at DU 14 and then passes through to the supra-scapular fossa to join ST 12.

One branch emerges behind the ear and enters the ear at Sanjiao 17 and then emerges in front of the ear to pass through SI 19 and ST7 on its way to the outer canthus.

It descends to the corner of the jaw near stomach 5 before ascending to the infra-orbital region and then on to meet UB1

It descends into the chest and meets the pericardium channel, it crosses the diaphragm and connects with the liver and gallbladder and then continues along the inside of the ribs down to emerging at the inguinal region.

It circles the genitals and runs posteriorly and meets the bladder channel at anterior sacrum and connects with DU1.

Then emerges on the buttock at gallbladder 30

Another branch descends from the supra-clavicular fossa to the axilla, runs along the side of the chest to GB22, 23 and 24.

It meets the liver channel at Liver 13 and then on to the hip joint to meet the previous branch at gallbladder 30. It descends down the lateral aspect of the thigh and knee before descending along the lateral aspect of the lower leg to just in front of the lateral malleolus.

It follows the dorsum of the foot along the groove between the fourth and fifth metatarsals to end on the lateral side of the tip of the fourth toe.

Another branch separates on the foot at GB 41 and runs between the first and second metatarsal bones to reach the medial tip of the big toe to link with the liver channel.

The GB channel connects with the following organs:

GB, liver

Meeting points:

ST 7, 8, 9, 12, SI12, 19, UB1, 11, 31, 32, 33, 34, PE1, SJ15, 20, 21, 22, LI13, DU1, 14, 20

I) LIV – primary

Originates at the lateral aspect of the dorsum of the big toe at Dadun LIV-1, and runs along the foot to Zhongfeng LIV-4, anterior to the medial malleolus.

Ascends along the medial aspect of the lower leg, intersecting the Spleen and kidney channels at Sanyinjiao SP-6, then continues to ascend anterior to the Spleen channel to an area 8 cun above the medial malleolus, where it crosses and continues posterior to the Spleen channel up to the knee and the medial aspect of the thigh.

Continues to the pubic region via Chingmen SP-12 and Fushe SP-13 where it encircles the genitals, then ascends to enter the lower abdomen where it intersects the Ren vessel at Qugu REN-2, Zhongji REN-3 and Guanyuan REN-4, Continues upwards to curve round the Stomach before entering the Liver and connecting with the Gall Bladder,

Crosses the diaphragm and spreads into the costal and hypochondriac region, Ascends along the neck and posterior aspect of the throat to the naso-pharynx, to link with the tissues surrounding the eye.

Ascends across the forehead to the vertex where it intersects with the Governing vessel at Baihui DU-20.

A branch

Descends from the eye system through the cheek and encircles the inner surface of the lips.

Another branch

Separates from the Liver, crosses the diaphragm and spreads into the Lung, meeting with P-1 – paired via the Jueyin division

The Liver channel connects with the following organs:

Liver, GB, LU, ST

Meeting points:

SP6, 12, 13, REN2, 3, 4, P1, DU20

